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wave. The given circuit gives distorted sine wave output.

2. What is the regulation of this stabiliser: 230 $\pm$ 1% or 230 $\pm$ 2%?

3. What frequency is the transformer designed to operate at?

Kishore Kumar Koneru
Design Engineer
Through e-mail

In response to Piyush Garg, the author Prof. K. Padmanabhan replies:

1. Capacitor C5 is 10 μ F but one can use any value between 4.7 μ F and 10 μ F.

2. Capacitor C6 (0.1 μ F) is not shown in the PCB layout. (It is to be connected externally across the output terminals of transformer X2.) It is a high-voltage paper or polyester capacitor to withstand AC mains voltage. It removes the AC ripples.

3. Fig. 3(c) shows how a sine wave opposite in polarity can be used to subtract from the mains supply when one wants to buck the mains voltage by an adjustable voltage generated by X2. I don't understand what point Mr Garg saw on the scope.

One is not supposed to see mains voltage (in this circuit) on the scope, as it may cause current flow through the common terminal of the scope to ground which may spoil the scope. But one can see the induced voltage on the transformer's secondary coil, keeping switch S2 open. This will vary with VR1 (10k) position in 'manual' mode of switch S1. It will change phase as VR1 is rotated from bottom to top. So, first, all testing has to be done with S2 open. *Regarding Kishore Kumar's letter:*

1. The waveform shown is just an illustration, with a more spaced pulse than the actual 5kHz. This is for one to note that the pulse does not swing down to ground or go negative. One can see that the capacitor charges during pulse-on time and discharges to send current through the winding of the transformer as seen from the exponential curve at the end of the pulse. One has to see that there is no high-frequency pulsing through the transformer, because then the core flux will swing positive and negative at this high a PWM frequency which the iron core will not be able to sustain. Also,

one can't obtain the phase change of 180 degrees needed for changeover from buck to boost. Thus one notes that the frequency of the current in the winding is predominantly 50 Hz with swings up and down on the sine shape of the wave. When we use 5 kHz (as in the actual program) – unlike as shown in the waveform illustration – the pulses will be too close to each other. The secondary output voltage after filtering by the other output capacitor becomes a smooth sine wave. So it is not that we provide an unsmooth waveform from this stabiliser.

2. The regulation is not a factor to be considered here because we are not controlling the voltage to be constant for variations in the load. Probably, what Mr Kumar means is the range of the voltage maintained during 170-250V input variations. That will be 5 per cent in this scheme, because we are sensing the voltage with a transformer of low voltage and offsetting that with a zener and evaluating how much more or less the input is. Accordingly, we adjust the pulse width to control the added secondary voltage.

It is possible to adjust it to between 215-225V, with a good 9V voltage transformer (X1), a zener diode (ZD1) and an ordinary diode in series so that at 220V input, the voltage at analogue pin 2 (RA0) of IC2 is 2.5V or half the supply voltage of the PIC. It also depends on the voltage of the secondary winding of the transformer – may be 55V or somewhat more. One can roughly calculate as indicated in the text.

3. As stated earlier, this is only a 50Hz voltage stabiliser transformer. The commercial voltage stabiliser transformer has only one primary winding, but a centre tap with both sides wound for 250V.

ERRATA

Please note that in the GW Instek AD published on page 162 of EFY's October 2010 issue, their distributor details have been wrongly mentioned. Readers may please note the correct distributor details as: Integrated Services & Consultancy, Bengaluru (An ESA Group Associate)